

A generating counterexample to the pure extension property in $\mathbb{M}_3$

Counterexample packet for `fa_banach_001`

11 August 2026

Status. Candidate full negative resolution, subject to human review. An explicit three-dimensional unital subspace $M \subset \mathbb{M}_3$ generates all of $\mathbb{M}_3$, yet a pure state on M has two distinct pure state extensions. The same phenomenon holds for the four-dimensional operator-system hull $M + M^*$.

Source question

Raphaël Clouâtre, *Restrictions of pure states to subspaces of C^* -algebras*, arXiv:2306.02365, *Journal of Functional Analysis* 289 (2025), no. 10, Article 111104, doi:10.1016/j.jfa.2025.111104 [1]. At the end of Section 2.3 (page 10 of the cached revised source PDF), after proving the result for $\mathbb{M}_2$, the paper states:

“In summary, any unital subspace of $\mathbb{M}_2$ has the pure extension property, while there are subspaces of $\mathbb{M}_4$ without this property [57]. We do not know what happens for subspaces in $\mathbb{M}_3$.”

$$F'(1) = -\frac{\sqrt{C}}{2} - 4 < F'(0) = 0 < F'(s_-).$$

In either case, we see that the maximum value of F is attained at a unique point in $[0, 1]$, as desired. $\square$

In summary, any unital subspace of $\mathbb{M}_2$ has the pure extension property, while there are subspaces of $\mathbb{M}_4$ without this property [57]. We do not know what happens for subspaces in $\mathbb{M}_3$.

Definitions

For a unital subspace M of a unital C^* -algebra B , a state on M is a unital contractive linear functional. It is *pure* when it is an extreme point of the state space. The space M has the *pure extension property in B* if every pure state on M has a unique state extension to B [1, §2]. A state on a unital subspace extends to a state on the ambient C^* -algebra: extend it norm-preservingly by Hahn–Banach, then use the standard fact that a unital norm-one functional on a C^* -algebra is positive.

Let E_{jk} denote the standard matrix units in $\mathbb{M}_3$, and let e_1, e_2, e_3 be the standard basis of $\mathbb{C}^3$.

Proof intuition

The construction creates a two-dimensional top eigenspace on which two different vector states cannot be distinguished, while two independent couplings to the third coordinate force the generated

C^* -algebra to be all of $\mathbb{M}_3$. The top eigenspace is cut out by $H = \text{diag}(1, 1, 0)$. Maximizing the expectation of H forces every extending density matrix to be supported on that eigenspace. The coupling matrices then have zero compression there, so all such extensions induce the same functional on the chosen subspace. Consequently that functional is not merely extreme but exposed. Nevertheless, the vector states at e_1 and e_2 remain distinct on $\mathbb{M}_3$.

Counterexample

Set

$$H = E_{11} + E_{22}, \quad X = E_{13} + E_{31}, \quad Y = E_{23} + E_{32},$$

and put

$$A = H + iX, \quad B = H + iY, \quad M = \text{span}_{\mathbb{C}}\{I, A, B\} \subset \mathbb{M}_3.$$

Theorem 1. *The subspace M has complex dimension 3 and $C^*(M) = \mathbb{M}_3$. The common restriction*

$$\varphi = \omega_1|_M = \omega_2|_M, \quad \omega_j(T) = \langle Te_j, e_j \rangle,$$

is an exposed, hence pure, state on M , but ω_1 and ω_2 are distinct pure state extensions of φ to $\mathbb{M}_3$. Consequently M does not have the pure extension property in $C^(M) = \mathbb{M}_3$.*

Moreover,

$$S = M + M^* = \text{span}_{\mathbb{C}}\{I, H, X, Y\}$$

is a four-dimensional operator system with $C^(S) = \mathbb{M}_3$, and the common restriction of ω_1, ω_2 to S is exposed and pure. Thus the failure persists even within the class of generating operator systems in $\mathbb{M}_3$.*

Proof. Generation. The matrices I, A, B are linearly independent because the E_{13} and E_{23} entries first force the coefficients of A and B to vanish. Hence $\dim_{\mathbb{C}} M = 3$. Because a generated C^* -algebra contains adjoints,

$$H = \frac{A + A^*}{2}, \quad X = \frac{A - A^*}{2i}, \quad Y = \frac{B - B^*}{2i}$$

belong to $C^*(M)$. Thus $E_{33} = I - H$ belongs to $C^*(M)$, and

$$XE_{33} = E_{13}, \quad E_{33}X = E_{31}, \quad YE_{33} = E_{23}, \quad E_{33}Y = E_{32}.$$

Products give

$$E_{13}E_{32} = E_{12}, \quad E_{23}E_{31} = E_{21},$$

as well as $E_{11} = E_{13}E_{31}$ and $E_{22} = E_{23}E_{32}$. All nine matrix units therefore lie in $C^*(M)$, so $C^*(M) = \mathbb{M}_3$.

The two extensions agree. For $j = 1, 2$,

$$\omega_j(I) = 1, \quad \omega_j(A) = 1, \quad \omega_j(B) = 1.$$

Thus the restrictions of ω_1 and ω_2 to M coincide; call the common state φ .

The common restriction is exposed. On the state space of M , consider the real affine functional

$$F(\psi) = \text{Re} \frac{\psi(A) + \psi(B)}{2}.$$

Let ψ be any state on M , and let ρ be a state extension to $\mathbb{M}_3$. Since X and Y are self-adjoint,

$$F(\psi) = \rho(H) \leq 1.$$

Suppose equality holds. Write $\rho(T) = \text{Tr}(DT)$ for a density matrix $D \geq 0$ with $\text{Tr } D = 1$. Then

$$1 = \rho(H) = D_{11} + D_{22}, \quad D_{33} = 0.$$

Positivity gives $|D_{j3}|^2 \leq D_{jj}D_{33} = 0$ for $j = 1, 2$. Hence the third row and column of D vanish, so

$$\rho(X) = \rho(Y) = 0.$$

It follows that $\psi(A) = \psi(B) = 1$, and therefore $\psi = \varphi$ on the basis I, A, B . Thus φ is the unique maximizer of F , so it is an exposed state and in particular is pure.

Nonuniqueness. Each ω_j is a vector state in the irreducible defining representation of $\mathbb{M}_3$, hence is pure. They are distinct because

$$\omega_1(E_{11}) = 1, \quad \omega_2(E_{11}) = 0.$$

They are therefore two distinct pure extensions of the pure state φ , proving the failure of the pure extension property.

Operator-system strengthening. The displayed adjoint formulas show

$$M + M^* = \text{span}_{\mathbb{C}}\{I, H, X, Y\} = S.$$

These four matrices are linearly independent and self-adjoint, and the generation argument already proves $C^*(S) = \mathbb{M}_3$. The restrictions of ω_1, ω_2 to S agree on the displayed basis. For a state σ on S , the affine functional $\sigma \mapsto \sigma(H)$ is at most 1; the density-matrix argument above shows that equality forces $\sigma(I, H, X, Y) = (1, 1, 0, 0)$. Hence the common restriction is again the unique maximizer, and is exposed and pure. Its two extensions remain distinct. $\square$

Verification notes

- The proof works in the strongest natural ambient interpretation: both M and S generate all of $\mathbb{M}_3$, so the ambient algebra equals $C^*(M) = C^*(S)$.
- No numerical approximation is used. The optional checker supplied in the packet's code directory verifies Hermiticity, the two vector-state restrictions, and that the generated matrix-unit span has complex dimension 9.
- The key purity step does not infer purity from purity of an extension. It proves directly that the common restriction is the unique maximizer of a real affine functional on the state space.
- Positivity is used only through the elementary density-matrix fact $D_{33} = 0 \Rightarrow D_{13} = D_{23} = 0$.

Novelty check and limitations

The cached source is the 31 October 2024 advanced revision. A bounded search performed on 11 August 2026 checked the arXiv record, the 2025 published version, and exact phrase/keyword combinations involving pure extension property, $\mathbb{M}_3$, operator systems, and the paper title. The published 2025 version still states the $\mathbb{M}_3$ question as open, and no later exact resolution was found.

Searches also inspected the authors' later operator-system work returned by these queries, but found no answer to this finite-dimensional question. This supports only a candidate-novel status; a specialist citation search remains appropriate before publication.

The theorem resolves the precise finite-dimensional question negatively. It does not address the paper's separate questions about the full noncommutative disc or Drury–Arveson algebras, nor its broader classification of uniquely extendible restrictions.

Human-review recommendation

High confidence in validity: the construction and proof reduce to explicit 3×3 matrix identities and a one-line positive-semidefinite argument. The primary review focus should be terminological—confirm that the paper's state convention for a nonself-adjoint unital subspace is the standard unital-contractive one used above—and novelty, because the result is surprisingly elementary. The operator-system strengthening independently avoids that convention and should be checked as the most robust form.

References

- [1] R. Clouâtre, *Restrictions of pure states to subspaces of C^* -algebras*, J. Funct. Anal. **289** (2025), no. 10, 111104; [arXiv:2306.02365](#).